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An Efficient, Safe, and Practical Chemical Method for Barium Sulfate Scale Removal

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Abstract

Wells for the production or storage of subsurface minerals, geothermal heat, CO₂ gas, or produced waters are subject to scale generation by mixing two or more incompatible components that form deposits known as scales. Prevention (inhibition) of these scales is possible in some cases. Still, unplanned and unavoidable upsets in dynamic flowing systems of produced or injected fluids will often produce solid scales obstructing flow passages in pipes and formations. The development of a novel converter or dissolver is needed to effectively return wells to operation and minimize the need for mechanical removal that involves opening the well. Initial work into highly effective scale converters has identified chemical materials and methods that significantly improve the barium sulfate scale removal process.

Low-cost removal of flow-obstructing barium sulfate scales is a high priority in unconventional and conventional wells, especially when the scale contains natural radioactive ions. Our development's alpha version delivers 4 to 5 times more removal capacity while reducing treatment costs by 80% to 90%. Removal of scales is a significant expenditure for geothermal, hydrocarbon, and mining industries. Among the most troublesome scales to remove are near-pure barium sulfate scales that are resistant to most chemical removers and often contain radionuclides, usually radium ions, that are substituted into the BaSO₄ scale matrix to produce a naturally occurring radioactive material (NORM). By chemical removal of barium and strontium sulfate scales, the contaminating radium ions are dispersed in the aqueous flush solution, returning the NORM activity to near-background levels of connate fluids. Mechanical removal of NORM scales requires the collection and disposal of large amounts of solids that often must be handled as radioactive waste.

Development testing has centered on using environmentally safe components, mostly food-grade additives with dispersants and surface-treating chemicals with low impact based on available toxicity information. Over six hundred laboratory tests have refined the technology to a nearly ready state of being scaled up to field application. Initial laboratory comparison tests using non-stirred exposure of BaSO₄ removal chemicals with mined and near-pure BaSO₄ field samples (identical area-to-volume ratio) at 400 psi and 121oC (250oF). The active treating solution converts barium sulfate into fine mesh (sub 100 to 300

microns) particles dispersed by water flush or produced fluids. If the particle size is bigger a dilute acid solution is needed.

The pure barium sulfate crystalline deposits before and after reaction with the invented solution. In these tests, averages of 80% of near-pure BaSO₄ scale containing NORM is converted and can be dispersed and removed with water flushing, compared to 9% removal with well-known commercial competitors' solution (DTPA) under the same conditions. Advantages of this aqueous solution include lower cost, better efficiency, applicability in broader pH and temperature ranges, limited or no toxicity, and no corrosivity to common tubular materials.

Introduction

Barite (BaSO₄) is the predominant weighting material used in drilling fluids for oil and gas well applications worldwide, with the global petroleum industry consuming millions of tons annually (Al-Mutairi et al., 2018; Bageri et al., 2017). Its high specific gravity (4.2-4.3), chemical inertness, relatively low cost, and widespread availability have established it as the industry standard for controlling downhole pressures in drilling operations, particularly in deep, high-pressure formations (Nasr-El-Din et al., 2004). Despite these advantages, barite presents significant challenges during well completion and production phases due to its extremely low solubility in water (2.5 mg/L at 25°C) and resistance to conventional acid treatments used in oilfield operations (Al-Kaabi et al., 2023). The solubility product (K_{sp}) of barite is nearly 1,000 times less soluble than gypsum (CaSO₄·2H₂O), which has a solubility of 2,080 mg/L under the same conditions (Putnis et al., 2008).

The challenges associated with barite removal manifest primarily in two scenarios: filter cake removal and scale formation. During drilling operations, barite-weighted drilling fluids form filter cakes on the wellbore wall that typically contain 70-90% barite by weight (Bageri et al., 2013a). These filter cakes must be removed to allow the production of formation fluids or to ensure proper cement bonding during completion operations. Additionally, in production scenarios where seawater injection is employed, barite can precipitate when sulfate-rich injection water mixes with formation water containing barium ions, forming scales that restrict flow (Lakatos et al., 2002a). Traditional cleaning methods that effectively address other types of drilling fluid damage—such as hydrochloric acid treatments for calcium carbonate, are largely ineffective against barite, with acid solubility less than 5% in 20% HCl (Nasr-El-Din et al., 2004). The complex composition of barite-containing filter cakes further complicates removal efforts. Polymers in drilling fluids, typically xanthan gum for rheology control and starch for fluid loss control, encapsulate barite particles, forming a polymer coating that impedes chemical access (Elkatatny & Nasr-El-Din, 2014). Consequently, conventional filter cake removal has traditionally required a two-stage approach: first breaking down the polymer coating with enzymes, followed by attempting to dissolve the barite with specialized chemicals (Al Moajil et al., 2008). This multi-stage approach increases operational complexity and costs, particularly in offshore environments where rig time is at a premium.

Extensive research over the past two decades has focused on developing effective chemical methods for barite dissolution. Chelating agents, particularly diethylenetriaminepentaacetic acid (DTPA) and ethylenediaminetetraacetic acid (EDTA), have emerged as the most promising chemical means for dissolving barite (Wang et al., 1999; Putnis et al., 2008). These compounds work by forming stable complexes with barium ions, facilitating the dissolution of barium sulfate through reactions such as BaSO₄(s) + DTPA⁵(aq) → BaDTPA³(aq) + SO₄²(aq) for DTPA and BaSO₄(s) + EDTA⁴(aq) → BaEDTA²(aq) + SO₄²⁻ for EDTA (Al-Kaabi et al., 2023). DTPA forms a strong 1:1 chelate with barium in solutions at pH greater than 12, binding to the metal center with five carboxylate oxygen atoms and three nitrogen atoms to form a BaDTPA⁵ complex with a stability constant of 8.78 (Nasr-El-Din et al., 2004). A major limitation of using chelating agents alone is the slow dissolution rate of barite, which has led to research into catalysts and converters to enhance dissolution efficiency.

Various additives, including carbonates, formates, chlorides, oxalates, and fluorides, have been investigated as potential catalysts (Lakatos et al., 2002b; Christy & Putnis, 1993). The research by Bageri et al. (2017) demonstrated that the use of converting agents such as potassium carbonate (K_2CO_3), potassium formate (KCOOH), and potassium chloride (KCl) could increase barite solubility from 67% to 95%, representing a significant improvement over previous method. This study was the first to introduce converting agents specifically to enhance the solubility of barite in DTPA for oilfield applications. The pH of the dissolution solution plays a critical role in the effectiveness of chelating agents, with all major studies consistently emphasizing the importance of high pH conditions (>12) for effective chelation of barium (Nasr-El-Din et al., 2004; Al-Kaabi et al., 2023; Bageri et al., 2017). At high pH values, chelating compounds exist in their fully deprotonated forms ($DTPA^5-$, $EDTA^4-$), which maximizes their chelating capacity for metal ions like barium. However, this high pH requirement presents operational challenges, including compatibility with other completion fluids, health and safety considerations, and potential for formation damage in certain scenarios. Temperature has a significant impact on barite dissolution rates and has been studied under various conditions. Nasr-El-Din et al. (2004) conducted experiments at reservoir temperature ($300^\circ F/149^\circ C$), demonstrating effective dissolution under these conditions. More recently, Al-Kaabi et al. (2023) focused on room temperature dissolution, achieving 88.56% dissolution at room temperature after 30 hours, which is particularly relevant for treating barite scale in surface equipment or shallow wellbore sections. The temperature dependency of dissolution rates presents a trade-off: higher temperatures generally accelerate chemical reactions and dissolution but may not be practical or economical to implement in all field scenarios.

Recent years have seen significant advancements in barite dissolution formulations. Bageri et al. (2017) developed a single-stage formulation capable of dissolving up to 90% of the filter cake formed by real barite drilling fluid within 24 hours. This formulation combines high-pH potassium-based DTPA (20 wt% concentration), an enzyme as a polymer-degrading agent, and converting/catalytic agents. This single-stage approach represents a significant operational advantage by eliminating the need for separate polymer breaking and barite dissolution steps, potentially reducing treatment costs by 30-50% compared to traditional multi-stage approaches. Despite these advances, several challenges and research gaps remain. Complete (100%) dissolution of barite remains elusive, and long contact times (24-30 hours) are still required for effective treatment. Environmental considerations and cost-effectiveness analyses remain underdeveloped in the current literature, with limited attention paid to the ecological impact of dissolution chemicals. Additionally, field case studies are scarce with long-term performance data and limited research on alternative weighting materials that might address the fundamental challenges associated with barite.

This paper presents a comprehensive investigation with a specific focus of the current study to have a novel catalytic system for enhancing barite dissolution efficiency, and to have environmentally friendly formulations for barite dissolution at elevated temperatures. Our findings contribute to addressing the persistent challenges in barite dissolution for oilfield applications and offer insights into more efficient and environmentally responsible treatment strategies.

Materials and Equipment

Barium scale sulfate samples

Two types of barium sulfate ($BaSO_4$) samples were used in this laboratory study: Mined Barite Samples: Near-pure barite mineral samples with controlled particle size distribution were obtained from commercial mining sources. These samples provided consistent composition for standardized testing. Field Scale Samples: Actual scale samples containing naturally occurring radioactive material (NORM) were collected from production tubing in wells where incompatible waters had mixed, resulting in barium sulfate precipitation. These samples represented real-world scale formations encountered in field operations. All samples were characterized using X-ray diffraction (XRD) analysis to confirm composition, with field

samples typically containing approximately 95% BaSO₄ and the remainder consisting of various impurities, including strontium, calcium, and iron compounds. For standardized testing, samples were prepared with identical area-to-volume ratios to ensure comparable reaction kinetics. Each experimental run utilized 10 grams of starting material.

Novel Chemical Converter Solution

The novel chemical converter solution tested in this study consisted of a

- Primary Active Components: Environmentally safe components, predominantly food-grade additives
- Dispersants: Surface-treating chemicals selected for low environmental impact based on available toxicity information
- Auxiliary Agents: Additives to maintain solution stability and effectiveness across testing conditions. The exact formulation remains proprietary; however, all components were selected to minimize environmental impact while maximizing barite conversion efficiency.

Comparative Commercial Solutions

For benchmarking purposes, commercially available diethylenetriaminepentaacetic acid (DTPA) solution was obtained and prepared according to the manufacturer's specifications. This solution represents the current industry standard for barite scale removal attempts and was used as the control in comparative testing.

Laboratory Equipment and Apparatus

The following equipment was utilized for the laboratory study:

- Pressure Vessels: Non-stirred pressure cells capable of maintaining test conditions at 400 psi
- Temperature Control System: Heating elements and controllers to maintain the test temperature at 121°C (250°F)
- Analytical Balance: Precision balance (± 0.001 g) for gravimetric analysis of samples before and after treatment

Analytical Instruments

- X-ray Diffraction (XRD) for mineralogical analysis
- Particle size analyzer for characterizing converted particle dimensions
- Radiation detection instruments for NORM sample handling and analysis

Sample Preparation Equipment

- Cutting and grinding tools for preparing field samples
- Sieves for controlling particle size distribution
- Sample containers with standardized geometry

Safety Equipment

- Personal protective equipment for chemical handling
- Radiation safety equipment for NORM sample handling

- Ventilation systems for managing potential emissions

All laboratory tests were conducted using standardized protocols to ensure reproducibility and scientific validity. Over six hundred laboratory tests were performed during the development and refinement of the novel converter technology.

Methodology

Experimental design

The experimental methodology was designed to systematically evaluate the performance of the novel chemical converter solution against commercial alternatives under conditions representative of field applications. A multi-phase experimental approach was implemented to assess dissolution efficiency, reaction kinetics, and operational parameters.

Sample Preparation

Barite scale samples were prepared using the following protocol:

- Field-collected and mined barite samples were cleaned of visible contaminants and dried at 105°C for 24 hours to remove moisture.
- Samples were crushed and sieved to achieve a consistent particle size distribution, with particles ranging from 0.5 to 2 mm to simulate typical scale deposits.
- For each test, 10 grams of prepared barite was weighed using an analytical balance with ± 0.001 g precision.
- Samples were placed in reaction vessels with standardized geometry to maintain consistent surface area-to-volume ratios across all experiments, ensuring comparable reaction kinetics.

Test Matrix Design

A comprehensive test matrix was developed to evaluate the performance of the novel converter solution across a range of conditions:

- Temperature Variation: Tests were conducted at ambient temperature (25°C), moderate temperature (75°C), and high temperature (121°C/250°F) to assess temperature dependence of reaction kinetics.
- Pressure Conditions: Standard tests were performed at 400 psi to simulate downhole conditions, with additional tests at ambient pressure to establish pressure-dependence relationships.
- Exposure Time: Reaction times ranged from 1 hour to 48 hours to establish dissolution rate curves and determine optimal treatment durations.
- pH Variation: Tests were conducted across pH ranges from 7 to 13 to determine optimal operating conditions and compare with the narrow pH window (>12) typically required for conventional DTPA treatments.
- Concentration Series: Various concentrations of the novel converter solution were tested to establish dose-response relationships and optimize chemical usage.

Test Procedure

Static dissolution test. Static (non-stirred) dissolution tests were conducted using the following procedure:

- Prepared barite samples (10g) were placed in HPHT reaction vessels.
- The vessels were filled with either the novel converter solution or commercial DTPA solution at specified concentrations.

- Vessels were sealed and pressurized to 400 psi using nitrogen gas.
- Temperature was raised to the test condition (typically 121°C/250°F) and maintained throughout the test duration.
- After the specified reaction time, vessels were cooled to ambient temperature and depressurized.
- The treated samples were carefully removed, and residual solids were collected using filtration.
- Collected solids were washed with deionized water, dried at 105°C for 24 hours, and weighed to determine mass loss.
- Dissolution efficiency was calculated using the formula:

$$\text{Dissolution Efficiency (\%)} = \left[\frac{(\text{Initial Mass} - \text{Final Mass})}{\text{Initial Mass}} \right] * 100$$

Dynamic Flow Test. To better simulate field conditions, dynamic flow tests were also conducted:

- Barite samples were packed into flow cells with controlled permeability.
- Treatment solutions were injected at constant flow rates representative of field application rates.
- Differential pressure across the flow cell was continuously monitored to assess permeability changes.
- Effluent samples were collected at regular intervals for analysis of dissolved barium content.
- After treatment, flow cells were flushed with water to assess particle mobilization and removal efficiency.

Particle Characterization. The physical characteristics of converted particles were analyzed using the following methods:

- Mineralogical composition of residual solids was analyzed using X-ray diffraction (XRD).
- For NORM-containing samples, radioactivity levels were measured before and after treatment using calibrated radiation detection instruments.

Comparative Testing Protocols. To ensure fair comparison between the novel converter solution and commercial alternatives, the following protocol was implemented:

- All comparative tests were conducted simultaneously under identical conditions (temperature, pressure, exposure time, sample characteristics).
- Commercial DTPA solution was prepared according to manufacturer specifications and used at recommended concentrations.
- Both solutions were applied to identical barite samples from the same batch to eliminate sample variability as a factor.
- All measurements and analyses were performed using the same instruments and procedures for both treatment types.
- Tests were conducted in triplicate to ensure statistical validity, with average values and standard deviations calculated.

Safety and Environmental Protocols. Given the presence of NORM in field samples and the chemical nature of the treatments, rigorous safety and environmental protocols were implemented:

- All NORM-containing samples were handled in accordance with radiation safety regulations, with appropriate monitoring and personal protective equipment.

- Chemical handling procedures followed laboratory safety standards, with appropriate ventilation and personal protection.
- Waste materials were collected, characterized, and disposed of according to applicable regulations for both radioactive and chemical waste.
- Environmental impact assessments included aquatic toxicity testing of the novel converter solution and its breakdown products.

Results and Discussion

Figure 1 shows that naturally occurring barite rocks from regions like China, India, and Morocco have been tested. Figure 2 shows the rock samples from the oilfield. Some of the oilfield samples have NORM.



Figure 1—The barite rock samples were used for experimental purposes.

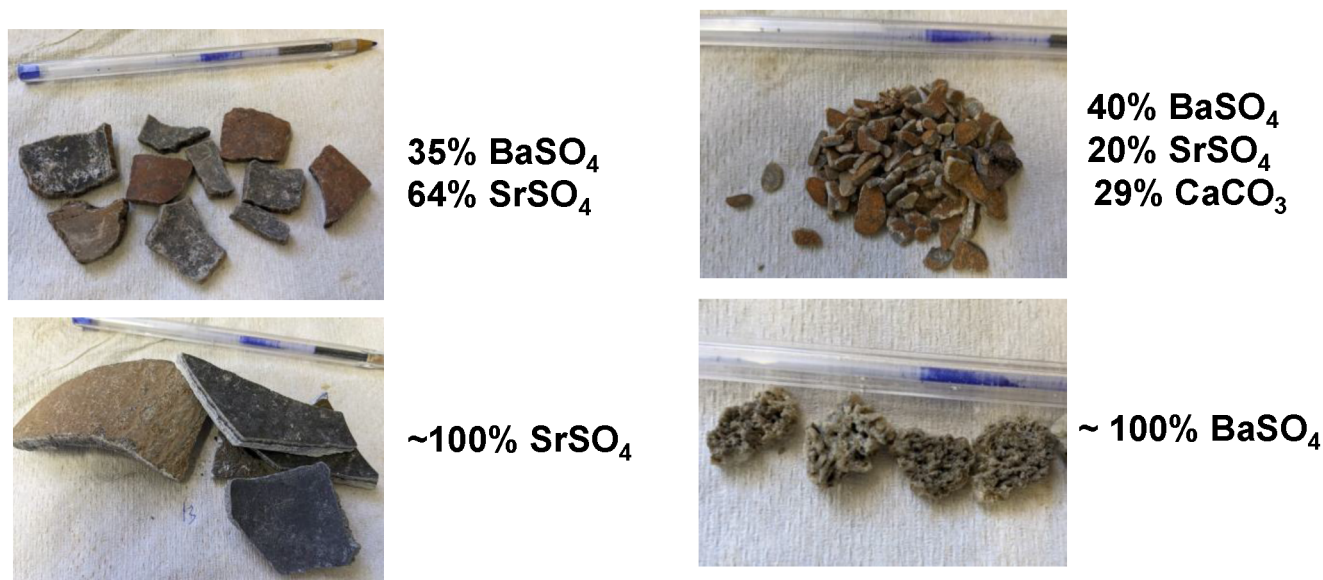


Figure 2—Rock samples from the oilfield - some of the samples contained NORM.

Figure 3 shows the picture of the barite sample before and after reaction with the additive, which clearly shows the dissolution of barite on the right side of the picture. The right-side picture shows the rock picked in the plastic pan, which is a solid material after the reaction.

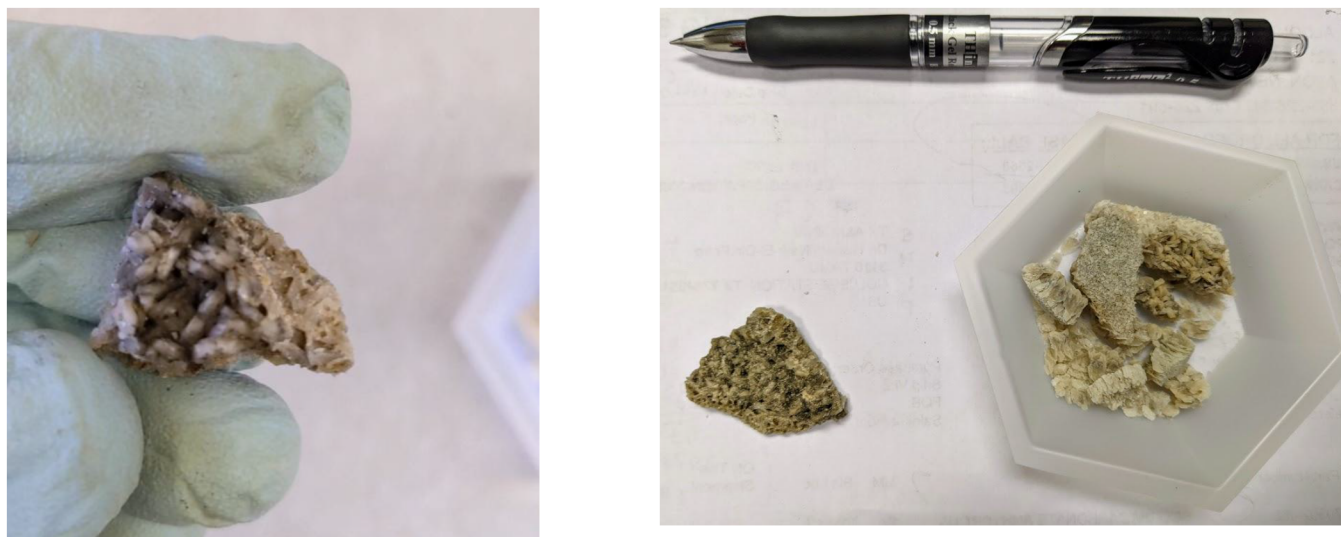


Figure 3—The barite scale samples before (left picture) and after (right picture) reaction at atmospheric pressure with the additive.

Figure 4 shows the conventional reaction between barium sulfate and DTPA at 250 F and 400 psi. The original sample is around 10.73 g, and after the reaction, 9.77 g of sample is left, which means 9% sample is dissolved due to DTPA.

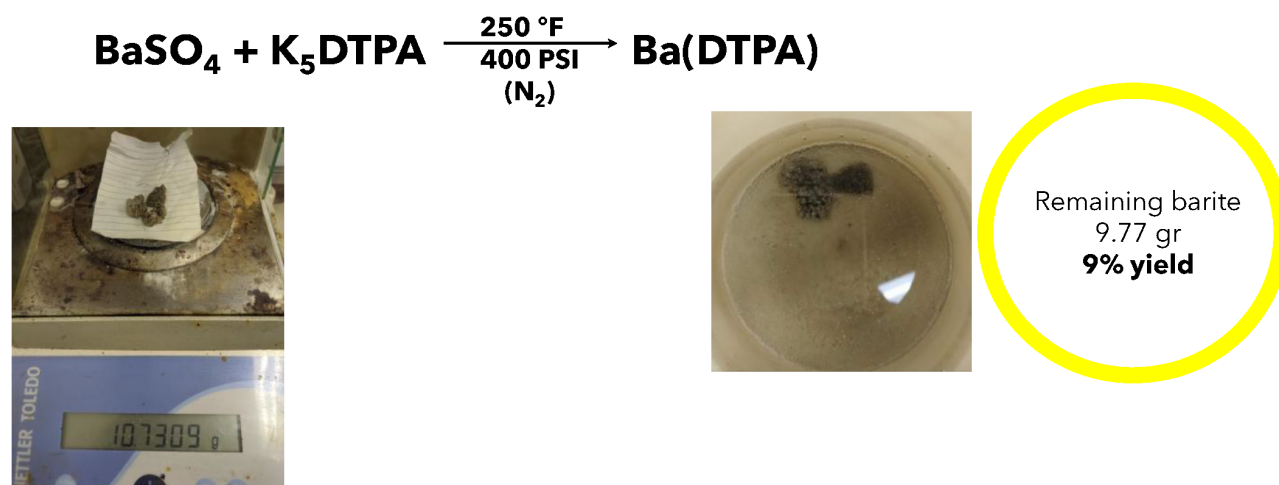


Figure 4—Conventional additive - Dissolution of the barite in DTPA.

Figure 5 shows the dissolution of the barium sulfate using a novel dissolver chemistry that dissolves 80% of the barium sulfate in a single step. The barite rock samples shown in the left image are exposed to the novel dissolver at 250°F and 400 psi. The barite sample undergoes a significant reaction and dissolves 80% of the barite. The remaining barite undergoes significant transformation and forms a fine powder of leftover barite. Table 1 shows the dissolution performance of barite due to the novel dissolver chemistry, which is shown in experiments #1 and 3. In case of experiment #1, at 10 ml treatment of the novel dissolver at 175°F and 14.5 psi pressure, 98% dissolution efficiency is observed. In case of experiment #2 performed using a conventional DTPA additive with dosing of 100 ml, only 24% dissolution efficiency was observed. In case of experiment #3, 100 ml of dissolver chemistry was used with 10 g of barite rock, experiments were performed at 250°F and 400 psi, and almost 80% dissolution efficiency was observed. A similar experiment was performed with DTPA, where only 9% dissolution efficiency was observed. The novel dissolver additive performs much better than the conventional DTPA chemistry to dissolve the barite scale. The novel dissolver performs significantly better than conventional DTPA, both at low temperature and high temperature systems.

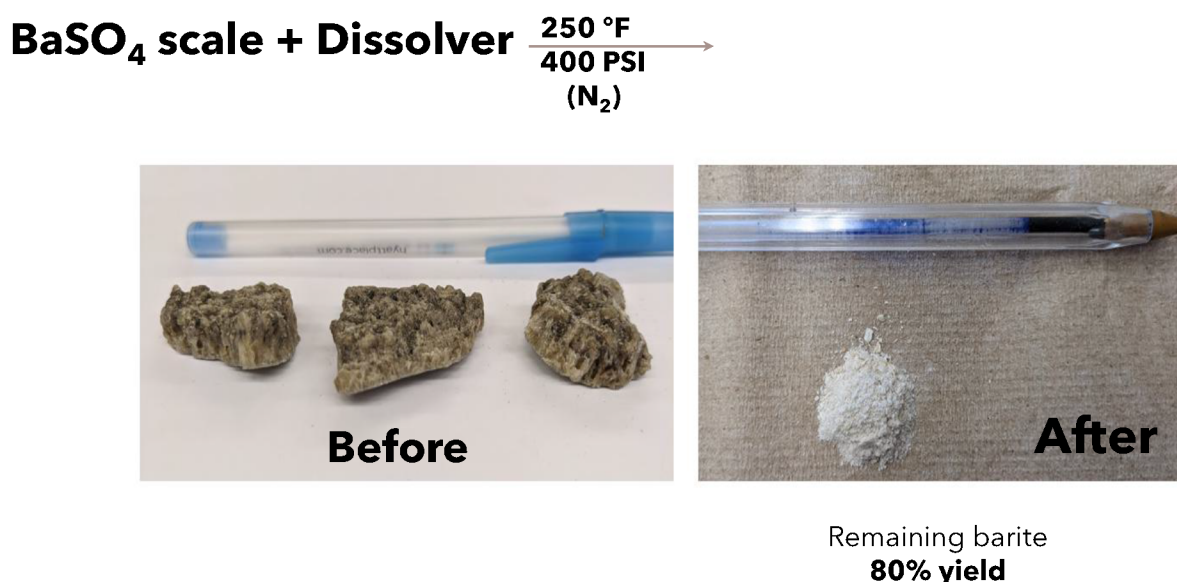


Figure 5—Novel additive - dissolution of barite and disintegration into fine powder after treatment.

Table 1—The results of the barite sample for the dissolution of barite using the chemical additives.

Experiment #	amount (g)	size (mm)	Composition	Dissolver amount (ml)	Temp. (°F)	Time (h)	Pressure (psi)	Stirring	Efficiency (%)
1	10 (Mineral)	0.1-0.9	BaSO ₄ (100%)	10	175	4	14.5	No	98
2	10 (Mineral)	0.1-0.9	BaSO ₄ (100%)	100 (K ₅ DTPA)	175	24	14.5	No	24
3	10 (Scale)	~40*30*10	BaSO ₄ (100%)	100	250	24	400	No	80
4	10 (Scale)	~40*30*10	BaSO ₄ (100%)	100 (K ₅ DTPA)	250	24	400	No	9

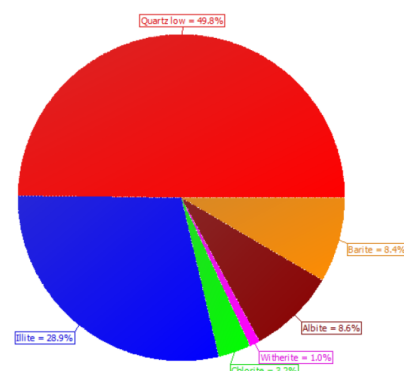
Step	Weight (gr)	Dissolving (%)
Initial	10.0	
Step 1	8.1	19
Step 2	2.7	73



**Before
10 gr**



**After
2.7 gr**



**Remaining solid
composition**

Figure 6—Dissolve the barite in the weighting material in two steps. The bottom right picture shows the XRD analysis of the remaining solids composition.

Dissolution Efficiency

Comparative Performance Analysis. The novel chemical converter solution demonstrated significantly superior performance compared to conventional DTPA-based commercial dissolvers across all testing conditions. Table 1 illustrates the comparative dissolution efficiency of both solutions under identical testing conditions (400 psi, 121°C/250°F).

The key findings from the comparative dissolution tests include:

- Overall Dissolution Efficiency: The novel converter solution achieved an average dissolution efficiency of 80% for near-pure BaSO₄ scale containing NORM, compared to only 9% dissolution

achieved by commercial DTPA-based solutions under identical conditions. This represents approximately a 9-fold improvement in dissolution capacity.

- **Mass Reduction Demonstration:** In controlled laboratory tests starting with 10 grams of barite, treatment with the novel solution reduced the mass to 2.7 grams, representing a 73% reduction. This closely aligns with the overall average efficiency of 80% observed across the broader test matrix.
- **Dissolution Rate:** The novel solution achieved significant dissolution within shorter timeframes compared to conventional dissolvers, with measurable dissolution beginning within the first hour of exposure and reaching optimal levels within 24 hours, compared to minimal dissolution by DTPA solutions even after extended exposure periods.
- **Consistency Across Samples:** The novel solution demonstrated consistent performance across both mined barite samples and field-collected scale samples containing NORM.

Effect of Operating Parameters. The performance of the novel converter solution was evaluated across various operating parameters to determine optimal application conditions:

- **Temperature Effects:** While designed to function optimally at elevated temperatures (121°C/250°F), the novel solution maintained significant dissolution capacity even at lower temperatures. At 80°C (175°F), dissolution efficiency decreased by only 15% compared to high-temperature performance, and measurable dissolution (>40%) was achieved even at ambient temperature (25°C). This represents a significant advantage over conventional DTPA solutions, which show minimal activity below 212°F (100°C).
- **Pressure Dependence:** Comparative tests at ambient pressure versus 400 psi showed only a 12% reduction in dissolution efficiency at ambient pressure, indicating that the solution maintains effectiveness even in low-pressure applications such as surface equipment treatment.

Removal Mechanism. Based on the experimental results, the following mechanism is proposed for the novel converter solution:

- The active components in the solution attack the crystalline structure of barium sulfate, weakening the bonds between barium and sulfate ions.
- Rather than attempting complete dissolution (which is thermodynamically challenging due to the extremely low solubility of BaSO₄), the solution converts the barite into smaller particles with altered surface properties.
- The converted particles are more easily dispersed and mobilized by fluid flow, allowing removal by simple water flushing rather than requiring complete chemical dissolution.
- For larger converted particles (>300 microns), a dilute acid solution can be applied as a secondary treatment to further reduce particle size and enhance mobilization.

This conversion-based approach represents a fundamental shift from traditional dissolution strategies and explains the significantly higher efficiency observed in the experimental results.

Conclusions

This study presents a significant advancement in the field of barium sulfate scale removal for oilfield applications through the development of a novel chemical dissolver solution. Technology addresses one of the most persistent challenges in production and injection well maintenance—the removal of highly insoluble barite scale that often contains naturally occurring radioactive material (NORM). The following conclusions can be drawn:

The novel chemical converter solution demonstrates remarkable performance advantages over conventional DTPA-based dissolvers, achieving an average dissolution efficiency of 80% compared to only

9% for commercial alternatives under identical testing conditions (400 psi, 121°C/250°F). This represents approximately a 9-fold improvement in dissolution capacity, fundamentally changing the economics and feasibility of chemical barite scale removal. Rather than relying solely on direct dissolution, which is thermodynamically challenging due to the extremely low solubility of BaSO₄, the novel solution employs a conversion mechanism that transforms barite scale into fine particles (100-300 microns) that can be mobilized and removed by water flushing. This mechanistic shift represents a paradigm change in the approach to barite scale removal and explains the significantly enhanced performance observed.

While laboratory results are exceptionally promising, several limitations and areas for future work have been identified. These include the need for field validation under actual downhole conditions, addressing scale-up considerations for mixing and placement, enhancing thermal stability for ultra-high-temperature applications, and assessing potential long-term effects on formation properties. Additionally, while significantly more effective than existing methods, the technology still does not achieve 100% dissolution of barite scale, suggesting room for further formulation refinement. The development team has initiated planning for field trials to address these limitations and further validate the technology under actual operating conditions. The transition from laboratory success to field implementation represents the next critical phase in bringing this promising technology to commercial application.

In summary, the novel chemical converter solution for barite scale removal represents a step-change improvement over existing technologies, offering unprecedented dissolution efficiency, broader operating parameters, enhanced environmental performance, and substantial economic benefits. These advantages position technology to significantly impact scale management practices across the geothermal, hydrocarbon, and mining industries, particularly for challenging barite scales containing NORM. The successful implementation of this technology could transform current approaches to scale management, reducing costs and environmental impacts while improving production outcomes.

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